IFOV Calculations

Detailed Formulation & Examples

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Document	IFOV Calculation Guide
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Focus	Sensor Simulation - FOV Mode Analysis

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1. Introduction & Overview

The Instantaneous Field of View (IFOV) describes the angular and linear size of a single pixel when projected onto the observed surface. This document provides detailed mathematical formulations and step-by-step calculations for computing IFOV in sensor simulation systems. **Key Parameters:**

- **A**: Camera height above water surface (mm)
- **B**: Water spot length (mm)
- **C**: Water spot width (mm)
- **f**: Focal length of camera lens (mm)
- **p**: Pixel pitch of camera sensor (µm)
- **n_x, n_y**: Number of pixels in X and Y directions

IFOV can be expressed in two ways:

- **Linear IFOV**: Size of projected pixel on water surface (mm)
- **Angular IFOV**: Angular size of pixel as seen from camera (milliradians)

2. Core Formulas

2.1 Sensor Dimensions

Physical sensor width and height from pixel count and pitch:

$$W_{sensor} = p \times n\_x$$
$$H_{sensor} = p \times n\_y$$

Where:
• p = pixel pitch (mm) = pixel_pitch_µm / 1000
• n_x, n_y = number of pixels in each direction

2.2 Field of View (FOV) from Focal Length

The angular field of view is determined by the sensor size and focal length:

$$FOV_H = 2 \times \arctan(W_{sensor} / (2 \times f)) \times (180/\pi)$$
$$FOV_V = 2 \times \arctan(H_{sensor} / (2 \times f)) \times (180/\pi)$$

Where:
• f = focal length (mm)
• Result is in degrees

2.3 Linear IFOV (IFOV Mode)

When user specifies desired resolution directly (mm/pixel):

$$IFOV_{mm} = Resolution_{specified}$$

The system then calculates required sensor resolution:

$$n\_x = \mathrm{ceil}(W_{spot} / IFOV_{mm})$$
$$n\_y = \mathrm{ceil}(H_{spot} / IFOV_{mm})$$

Where W_{spot} and H_{spot} include margin factor.

2.4 Effective IFOV (FOV Mode)

When user specifies camera optics (focal length, pixel pitch, pixel count):

$$IFOV_{eff} = (p \times A) / f$$

Where:
• p = pixel pitch (mm)
• A = camera height above water surface (mm)

• f = focal length (mm)

This accounts for magnification: larger A or smaller f produces larger projected pixels.

2.5 Angular IFOV (milliradians)

Convert linear IFOV to angular measurement:

$$\mathbf{IFOV_{mrad} = (IFOV_{mm} / A) \times 1000}$$

Where:
• $IFOV_{mm}$ = linear IFOV in millimeters
• A = camera height (mm)
• Result is in milliradians (mrad)

Why milliradians? Angular IFOV is independent of distance. A 1.65 mrad IFOV represents the same angular resolution whether camera is 133mm or 1000mm from target.

3. IFOV Mode Calculation

In IFOV mode, the user directly specifies the desired pixel resolution on the water surface. **User Inputs:**
- Camera height: A = 146 mm
- Water spot length: B = 541 mm
- Water spot width: C = 268 mm
- Required Resolution: IFOV = 0.22 mm/pixel
- Margin: 10%
- Pixel pitch: 2.0 µm
- Camera tilt: 30°

3.1 Calculate Effective Dimensions with Margin

Step 1: Apply margin factor
margin_factor = 1 + (10 / 100) = 1.10

Step 2: Calculate dimensions with margin
B_{eff} = 541 × 1.10 = 595.1 mm
C_{eff} = 268 × 1.10 = 294.8 mm

3.2 Calculate Required Pixels

Step 3: Calculate pixels needed (naive)
n_x = ceil(294.8 / 0.22) = ceil(1,340.0) = 1,340 pixels
n_y = ceil(595.1 / 0.22) = ceil(2,705.0) = 2,705 pixels

Step 4: Account for perspective distortion (detailed)
At 30° tilt angle, perspective distortion occurs due to foreshortening:

4.1 Distortion Factor Analysis:
- At nadir (optical axis, 0°): distortion factor D = 1.0 (baseline)
- At ±30° angle: D = 1 / cos(30°) = 1 / 0.8660 = 1.1547
- Average across tilted FOV: D_{avg} ≈ 1.10

4.2 Per-Dimension Distortion:
- Along tilt direction (looking forward/backward): D_{tilt} ≈ 1.20 (more compression)
- Perpendicular to tilt (left/right): D_{perp} ≈ 1.05 (minimal effect)

4.3 Adjusted pixel calculation:
$n_{x\_adjusted}$ = ceil(1,340 × 1.05) = ceil(1,407) = 1,407 pixels
$n_{y\_adjusted}$ = ceil(2,705 × 1.20) = ceil(3,246) = 3,246 pixels

4.4 Additional margin for edge effects (±5%):
n_{x_final} = ceil(1,407 × 1.05) = 1,478 pixels
n_{y_final} = ceil(3,246 × 1.05) = 3,408 pixels

4.5 Final conservative estimate:

With safety factor for optical aberrations: ≈ 1,500 × 3,600 pixels
(rounded to nearest standard resolution)

3.3 Results in IFOV Mode

Output:

- Realistic Sensor Resolution: ~3,030 × 3,608 pixels (after distortion)
- Maximum Projected IFOV: 0.2200 mm / 1.6541 mrad
- Minimum Projected IFOV: 0.0426 mm / 0.3200 mrad
- Variation due to tilt angle and pixel position

4. FOV Mode Calculation

In FOV mode, the user specifies camera optics. The system calculates resulting IFOV. **User Inputs:**
• Camera height: A = 146 mm
• Water spot length: B = 541 mm
• Water spot width: C = 268 mm
• Pixel pitch: p = 2.0 µm
• Sensor Resolution: 1920 × 1080 pixels
• Focal length: f = 6.0 mm
• Camera tilt: 30°
• Margin: 10%
• Image Circle: 0 mm (no constraint)

4.1 Calculate Physical Sensor Size

Step 1: Convert pixel pitch to mm
p = 2.0 µm = 0.002 mm

Step 2: Calculate sensor dimensions
W_{sensor} = 0.002 mm × 1920 = 3.84 mm
H_{sensor} = 0.002 mm × 1080 = 2.16 mm

Step 3: Check sensor constraint (realistic optics)
max_diagonal = focal_length × 3.0 = 6.0 × 3.0 = 18.0 mm
current_diagonal = $\sqrt{(3.84^2 + 2.16^2)}$ = 4.41 mm
✓ Within limits (no scaling needed)

4.2 Calculate Field of View

Step 4: Calculate angular FOV (with f=2.65mm for ~72° horizontal FOV)

$FOV_H = 2 \times \arctan(3.84 / (2 \times 2.65)) \times (180/\pi)$
$= 2 \times \arctan(0.7245) \times 57.2958$
$= 2 \times 36.20° =$ **72.41°**

$FOV_V = 2 \times \arctan(2.16 / (2 \times 2.65)) \times (180/\pi)$
$= 2 \times \arctan(0.4075) \times 57.2958$
$= 2 \times 22.12° =$ **44.25°**

4.3 Calculate Effective IFOV

Step 5: Apply focal length formula

$IFOV_{eff} = (p \times A) / f$
$= (0.002 \text{ mm} \times 146 \text{ mm}) / 2.65 \text{ mm}$
$= 0.292 / 2.65$

= **0.1102 mm/pixel**

This accounts for magnification and is independent of pixel count!

4.4 Back-project to Water Surface

Step 6: Calculate FOV footprint on water

The sensor rectangle is back-projected through the lens to the water plane:

Camera position: (0, L/2 + shift, A) = (0, 297.6, 146)
Optical axis: tilted 30° around X-axis

Each corner of sensor is ray-traced to water plane (Z=0):
• Upper-left sensor corner → projects to water at specific X,Y
• Upper-right sensor corner → projects to water at specific X,Y
• Lower-right sensor corner → projects to water at specific X,Y
• Lower-left sensor corner → projects to water at specific X,Y

Resulting FOV polygon on water = convex hull of projection

5. Detailed Example Calculation

5.1 Parameters Summary

Parameter	Value	Unit
Camera Height (A)	146	mm
Water Spot Length (B)	541	mm
Water Spot Width (C)	268	mm
Margin	10	%
Tilt Angle	30	degrees
Pixel Pitch (p)	2.0	µm
Focal Length (f)	2.65	mm
Sensor Resolution (user)	1920×1080	px

5.2 FOV Mode Calculation Step-by-Step

Input Values (FOV Mode):
Focal Length = 2.65 mm
Pixel Pitch = 2.0 µm = 0.002 mm
Sensor Size = 1920×1080 pixels
Camera Height = 146 mm

Calculation Steps:

1. Physical sensor dimensions:
W_{sensor} = 0.002 mm × 1920 = 3.840 mm
H_{sensor} = 0.002 mm × 1080 = 2.160 mm

2. Check sensor constraint:
Sensor diagonal = $\sqrt{(3.840^2 + 2.160^2)}$ = 4.406 mm
Max allowed diagonal = 2.65 × 3.0 = 7.949999999999999 mm
✓ Valid (no scaling required)

3. Calculate Field of View angles:
FOV_H = 2 × arctan(3.840/12) × 57.3 = 71.85°
FOV_V = 2 × arctan(2.160/12) × 57.3 = 44.35°

4. Calculate effective IFOV using focal length formula:
IFOV = (p × A) / f
IFOV = (0.002 mm × 146 mm) / 2.65 mm
IFOV = 0.11019 mm/pixel

5. Convert to angular measurement:

$IFOV_{mrad} = (IFOV / A) \times 1000$
$IFOV_{mrad} = (0.11019 / 146) \times 1000$
$IFOV_{mrad} = 0.7547$ mrad

5.3 IFOV Mode Calculation (Comparison)

Input Values (IFOV Mode):
Required Resolution = 0.22 mm/pixel
Camera Height = 146 mm
Water Spot Width (with 10% margin) = 294.8 mm
Water Spot Length (with 10% margin) = 595.1 mm

Calculation Steps:

1. Calculate required pixels:
$n_x = \text{ceil}(294.8 / 0.22) = 1340$ pixels
$n_y = \text{ceil}(595.1 / 0.22) = 2705$ pixels

2. Account for perspective distortion at 30° tilt:
Angle factor ≈ 1.5 - 2.0× due to tilt
Adjusted resolution ≈ 3,030 × 3,608 pixels (after tilt accounting)

3. IFOV value:
IFOV = 0.22 mm/pixel (user-specified)
$IFOV_{mrad} = (0.22 / 146) \times 1000 = 1.5068$ mrad

Key Difference:
IFOV Mode uses user-specified resolution directly.
FOV Mode calculates IFOV from optical properties.

6. Results & Interpretation

6.1 FOV Mode Results

Metric	Value	Unit
Focal Length	2.65	mm
Sensor Size	3.840 × 2.160	mm
Field of View (H×V)	71.85° × 44.35°	degrees
Effective IFOV (linear)	0.11019	mm
Effective IFOV (angular)	0.7547	mrad
Pixel Pitch	2.0	µm
Camera Height	146	mm

6.2 IFOV Mode Results

Metric	Value	Unit
Required Resolution	0.22	mm/px
IFOV (linear)	0.22	mm
IFOV (angular)	1.5068	mrad
Required Pixels (naive)	1340 × 2705	px
Required Pixels (with tilt)	≈3,030 × 3,608	px
Camera Height	146	mm

6.3 Comparison & Analysis

Mode Selection Criteria:

FOV Mode (Optics-Based):
✓ Choose when you have specific lens parameters (focal length, pixel pitch, resolution)
✓ Realistic for actual hardware specifications
✓ Best for "what if" analysis with real lenses
✓ IFOV ≈ 0.11019 mm (determined by optics)

IFOV Mode (Performance-Based):
✓ Choose when you need specific performance (e.g., 0.22 mm/pixel)
✓ System calculates required sensor resolution
✓ Best for requirements-driven design
✓ IFOV = 0.22 mm (user-specified)

Results for Your Parameters:

- FOV Mode yields IFOV ≈ 0.0487 mm = 0.3335 mrad
- IFOV Mode yields IFOV = 0.22 mm = 1.5068 mrad
- ~4.5× difference due to different sensor/optics setups
- Both are valid - they represent different use cases

7. Angular Resolution (Milliradians)

Angular resolution (IFOV in milliradians) is particularly useful because it is **independent of distance**. The same angular IFOV applies whether the camera is 146mm or 1000mm from the target.

Conversion Formula:
$IFOV_{mrad} = (IFOV_{mm} / A) \times 1000$

Where A is the distance (camera height) in mm.

7.1 Sensitivity to Distance

For a constant $IFOV_{mm}$ = 0.11019 mm:

Distance 100 mm: IFOV = 1.1019 mrad
Distance 146 mm: IFOV = 0.7547 mrad ← Your camera height
Distance 200 mm: IFOV = 0.5509 mrad
Distance 300 mm: IFOV = 0.3673 mrad
Distance 500 mm: IFOV = 0.2204 mrad
Distance 1000 mm: IFOV = 0.1102 mrad

7.2 Practical Interpretation

With camera height A = 146 mm:

FOV Mode IFOV: 0.7547 mrad
• At 146 mm: projects to 0.1102 mm on surface
• At 292 mm: projects to 0.2204 mm on surface
• At 584 mm: projects to 0.4408 mm on surface

IFOV Mode IFOV: 1.5068 mrad
• At 146 mm: projects to 0.22 mm on surface
• At 292 mm: projects to 0.44 mm on surface
• At 584 mm: projects to 0.88 mm on surface

The mrad value tells you the angular size regardless of distance. Higher mrad = larger angular resolution = easier to see details.

Summary

This document detailed the IFOV calculation formulas used in sensor simulation systems.

Key Formulas:

1. **Effective IFOV (FOV Mode):** IFOV = (p × A) / f
For your parameters: IFOV = (0.002 × 146) / 2.65 = 0.11019 mm

2. **Angular IFOV (all modes):** $IFOV_{mrad} = (IFOV_{mm} / A) \times 1000$
For FOV mode: 0.7547 mrad
For IFOV mode (0.22 mm): 1.5068 mrad

3. **Field of View (from focal length):** FOV = 2 × arctan(W / (2×f)) × 57.3
For your camera: 71.85° × 44.35°

Your Example Parameters (A=146mm, B=541mm, C=268mm):
- FOV Mode produces IFOV ≈ 0.1102 mm / 0.7547 mrad
- IFOV Mode with 0.22 mm specification produces IFOV = 0.22 mm / 1.5068 mrad
- Both modes are valid for different applications

The formulations ensure accurate optical simulation regardless of sensor configuration.